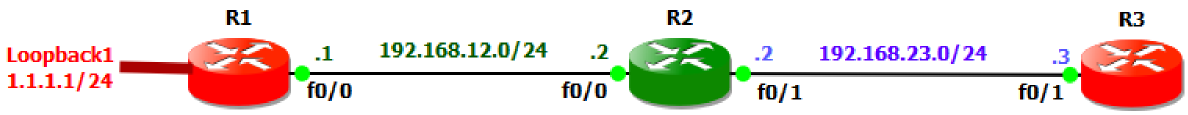
EIGRP Basic Configuration:

Let us do basic configuration of EIGRP protocols on cisco routers. Wildcard mask is 32 bits long. It is inverted subnet masks, with the zero bits indicating that the corresponding bit position must match the same bit position in the IP address. The one bits indicate that the corresponding bit position does not have to match the bit position in the IP address.

To form a neighborship, EIGRP has these requirements: **1)** Interface's primary IP addresses must be on the same subnet.**2)** Connected interface must not be passive.**3)** Routers must use the same AS number.**4)** Must pass authentication. **5)** K-values must match.



Commands	**Description**
R1(config)#router eigrp 1 R1(config-router)#network 192.168.12.0 R1(config-router)#network 1.1.1.0	Enter EIGRP mode Advertise 192.168.12.0 network Advertise 1.1.1.0 network
R1#show run | sec eigrp router eigrp 1 network 1.0.0.0 network 192.168.12.0	To verify by running configuration EIGRP with AS 1 Its convert 1.1.1.0 network class A
R1(config)#router eigrp 1 R1(config-router)#no network 1.1.1.0 R1(config-router)#network 1.1.1.0 0.0.0.255	Enter EIGRP mode Remove network Advertise again with wild card
R1#show run | sec eigrp router eigrp 1 network 1.1.1.0 0.0.0.255	Verify by running configuration EIGRP with AS 1 This time it's not converted
R1(config)#router eigrp 1 R1(config-router)#no network 1.1.1.0 0.0.0.255 R1(config-router)#network 1.1.1.0 255.255.255.0	Enter EIGRP mode Remove network With subnetmask not with wildcard
R1#show run | sec eigrp router eigrp 1 network 1.1.1.0 0.0.0.255	Verify by running configuration EIGRP with AS 1 IOS converted subnetmask to wildcard
R1(config)#router eigrp 1 R1(config-router)#network 0.0.0.0	Enter EIGRP mode with AS 1 Advertise all networks
R2(config)#router eigrp 100 R2(config-router)#network 0.0.0.0	Enter EIGRP mode with AS 100 Advertise all networks, no neighbor adjacency because of different AS
R1(config)#router eigrp 1	Enter EIGRP mode with AS 1

R1(config-router)#network 0.0.0.0 R1(config-router)# metric weights tos 1 1 1 1 1	Advertise all networks Changing Metric Weights Again no neighbor adjacency, K value mismatch
R2(config)#router eigrp 1 R2(config-router)#network 0.0.0.0	Enter EIGRP mode with AS 1 Advertise all networks

EIGRP Commands:

Commands	**Description**
R1(config-router)#auto-summary	Enable auto summarization feature
R1(config-router)#no auto-summary	Disable auto summarization feature
R1(config-router)# metric weights tos k1 k2 k3 k4 k5	Adjusting the EIGRP Metric Weights
R1(config-router)# metric maximum-hops <1-255>	Advertise greater than hops
R1(config-router)#maximum-paths <1-32>	Set the maximum equal paths
R1(config-router)#variance <1-128>	Control unequal load balancing
R1(config-if)# ip hello-interval eigrp <asn> <interval>	Changing EIGRP hello interval
R1(config-if)# ip hold-time eigrp <asn> <interval>	Changing EIGRP hold time interval
R1#show ip eigrp neighbors	Display the neighbor table in brief
R1#show ip eigrp neighbors detail	Display the neighbor table in detail. To verify the neighbor is configured as stub router
R1#show ip eigrp interfaces	Display info about all EIGRP interfaces
R1#show ip eigrp interfaces s0/0	Display info EIGRP interface
R1#show ip eigrp interfaces 20	Display info EIGRP interfaces AS 20
R1#show ip eigrp topology	Displays the topology table
R1#show ip eigrp traffic	Displays EIGEP different packets
R1#show ip route eigrp	Display EIGRP route from routing table
R1#debug eigrp fsm	Displays the events related to FSM
R1#debug eigrp packet	Displays EIGRP event packets
R1#no debug eigrp fsm	Turn off FSM debug
R1#no debug eigrp packet	Turn off EIGRP packets debug